Mobile Product Analytics Portfolio • Automated Daily Briefing • *2026-08-22*

■ **Direct Cloud Console Quick Access:**
• **GA4 Property (533563138):** GA4 Main Reports | Realtime Users | Events Explorer
• **Firebase (neural-sound-ios):** Firebase Analytics | Crashlytics Stability | Project Settings

Portfolio Executive Summary & Health Dashboard

Side-by-Side Application Performance Comparison

Metric / KPI	Music Separation iOS (com.neuralsound.mss)	Vocal Cleaner (com.neuralsound.audioCleaning)	Portfolio Total / Avg
Yesterday Active Users (AU)	143	3	**146**
Yesterday New Users (NU)	107	0	**107**
Yesterday Sessions	169	17	**186**
30-Day Active Users	841	27	**868**
Lifetime Active Users	1,056	27	**1,083**
Yesterday Engagement Rate	84.6%	88.2%	**84.9%** (Weighted Avg)
Yesterday Avg Session Duration	333.1s (~5.5 min)	3,531.1s (~58.8 min)	**625.3s** (Weighted Avg)
Crash-Free User Rate (24h)	100.0%	66.67%	**99.3%** (Portfolio Avg)
Primary Horizon Trajectory	**Hyper-Growth (+255% 7D Surge)**	**Stagnant / Friction-Bound**	**Single-App Growth Driven**
Key Strategic Risk	Telemetry Event Bloat & Rollout Degradation	Critical Crash Rate & Zero Organic Acquisition	Concentration Risk (App & Geo)

Cross-App Strategic Portfolio Synthesis

1. **Portfolio Acquisition Asymmetry & Concentration Risk**:
   The Neural Sound portfolio is currently single-threaded in its growth. **Music Separation iOS** accounts for **97.9% of active daily users** and **100% of new acquisitions** over the past 24 hours. Conversely, **Vocal Cleaner** exhibits zero organic user acquisition (`0 New Users` yesterday, 7 cumulative lifetime new users in 7 days), rendering it entirely dependent on legacy retained users.
2. **Telemetry Infrastructure & GA4 Cost Bloat**:
   A major architectural telemetry flaw exists in **Music Separation iOS**. Continuous UI control elements (sliders for Pitch and Tempo) fire un-debounced analytical events (`player_pitch_changed` averaged **570.6 events/user** across 10 users). This artificially inflates event volumes, distorts processing pipelines, and risks exceeding standard GA4 event rate limits.
3. **Monetization Engine vs. Top-of-Funnel Bottlenecks**:
   Vocal Cleaner demonstrates high monetization intent and power-user engagement (average session duration of ~58.8 minutes, multiple subscription renewals, and high paywall interaction). However, extreme StoreKit transaction friction (`purchase_failed` count = 5, `purchase_storekit_failed` count = 2) combined with a severe stability deficit (**66.67% crash-free rate**) completely suffocates revenue realization.
4. **Staged Rollout Degradation & Quality Assurance Alerts**:
   Music Separation iOS version `1.3.08` (currently in staged rollout to 5 users) shows a steep drop in engagement rate (**53.8%** vs. baseline `1.3.07` at **86.8%**). Immediate investigation is required before expanding the rollout percentage.

Deep-Dive Application Analysis

1. Music Separation iOS (`com.neuralsound.mss`)

App Header & Health Score

- **GA4 Property ID**: `533563138` | **Stream ID**: `14402502632`
- **Overall Health Score**: **8.2 / 10**
- **Score Rationale**: Strong viral organic growth acceleration (+255% DAU expansion over 7 days), near-perfect onboarding completion (~97.2%), and flawless production stability (100% crash-free user rate on primary build `1.3.07`). The score is penalized due to **severe telemetry debouncing bugs** (event spamming on UI sliders), **geographic hyper-concentration** in South Korea (76.9% of traffic), and an **engagement drop on the newly staged build `1.3.08`**.

■ Multi-Horizon Growth & Historical Scale Comparison

Horizon	Active Users (AU)	New Users (NU)	Total Sessions	Engagement Rate	User Base Growth Velocity Analysis
Yesterday (Latest 24h)	143	107	169	84.6%	**Parabolic Surge**: NU accounts for 74.8% of daily active users.
Past 7 Days	419	311	595	84.5%	**Velocity Expansion**: 74.2% of weekly users joined within the last 7 days.
Past 30 Days	841	708	1,648	85.6%	**Scale Acceleration**: 84.2% of total 30-day cohort acquired within 30 days.
Past Year (365 Days)	1,056	933	2,487	84.8%	**Early-Stage Life Cycle**: App has reached maturity inside the last 12 months.
All-Time / Lifetime	1,056	933	2,487	84.8%	**Lifetime Total Baseline**.

```
[Lifetime AU: 1,056] → [30D AU: 841 (79.6%)] → [7D AU: 419 (39.7%)] → [24h AU: 143 (13.5%)]
```

Trajectory & Retention Analysis

- **Growth Velocity**: Acquisition velocity is accelerating rapidly. Yesterday's volume (107 new users) represents **34.4% of the entire 7-day new user cohort** and **15.1% of all-time new users**.
- **Retention Dynamics**: While short-term engagement is exceptional (>84%), comparison of Lifetime AU (1,056) vs. 30D AU (841) indicates that the app is early in its growth lifecycle, with early cohort retention remaining unproven beyond 30 days.

Old vs. New User Cohort Deep-Dive

User Cohort Type	Active Users	Total Sessions	Avg Session Duration	Engagement Rate	Screen / Page Views	Key Architectural Observation
New Users	114	112	347.8 sec (~5.8 min)	94.6%	229	**Exceptional Onboarding**: 94.6% engagement rate proves immediate value payoff.
Returning Users	41	52	332.9 sec (~5.5 min)	69.2%	39	**Engagement Decay**: Returning users show a 25.4% drop in engagement rate.
(not set)	0	5	6.5 sec	20.0%	1	Background/Ghost sessions or delayed attribution payloads.

Cohort Insights

1. **New User Onboarding Efficiency**: New users spend **347.8 seconds** per session with a **94.6% engagement rate**. This confirms that initial audio separation flows deliver fast time-to-value.
2. **Returning User Utility Shortfall**: Returning users generate only **0.75 screen views per session** (39 page views across 52 sessions), compared to **2.04 screen views per session** for new users. Returning users quickly launch the core separation player without exploring auxiliary features.

7-Day Velocity & Trajectory

Date	Active Users	New Users	Sessions	Engagement Rate	Contextual Trend Analysis
2026-08-15	30	4	46	82.6%	Baseline operation period.
2026-08-16	28	7	38	84.2%	Flat volume baseline.
2026-08-17	36	14	45	80.0%	Early acquisition movement (+100% NU WoW).
2026-08-18	57	36	71	83.1%	Tipping point: New user acquisition surge begins.

2026-08-19	94	78	120	**90.8%**	Viral adoption peak: Korean market campaign activation.
2026-08-20	93	65	113	82.3%	Mid-week consolidation.
2026-08-21	**143**	**107**	**169**	**84.6%**	**All-Time Peak**: Daily active user volume increases +376% vs 08-15.

User Funnel Diagrams

Onboarding & First-Journey Funnel

```
[109 Users] first_open ■ ▼ (100.0% Conversion) [111 Users] onboarding_viewed ■ ▼ (100.0% Step Progression) [111 Users] onboarding_step_viewed (449 events, ~4.0 steps/user) ■ ▼ (99.1% Tap Through) [110 Users] onboarding_next_tapped ■ ▼ (100.0% Screen Load) [117 Users] login_screen_viewed ■ ▼ (94.0% Auth Rate) [110 Users] login_succeeded ■ ▼ (96.4% Completion Rate) [106 Users] first_journey_login_completed ■■■ Overall Onboarding Conversion: 97.2%
```

- **Funnel Analysis**: Onboarding is optimized. **97.2% of users** who open the app complete the authentication and first-journey onboarding flow.

Core Audio Import & Processing Execution Funnel

```
[105 Users] plus_button_tap / [93 Users] first_journey_add_separation_clicked ■ ■■■■ [78 Users] import_files_app_tap ■■■ ■ ■■■■ [141 Unique Import Intent Users] ■■■■ [63 Users] import_photos_tap ■■■■■ ■ ▼ (37.5% Selection Drop-Off) [53 Users] import_started (105 events) ■ ▼ (88.7% Processing Success) [47 Users] separation_play_started (571 events) ■ ▼ (100% Player Open Expansion) [72 Users] separation_player_opened (199 events)
```

- **Funnel Drop-off Diagnostic**:
- **Critical Bottleneck**: A **37.5% conversion drop-off** occurs between selecting an import source (`import_files_app_tap` / `import_photos_tap`) and triggering `import_started` (78 + 63 = 141 tap intents vs. 53 actual starts).
- **Root Cause**: File format validation failures, iOS Document Picker modal cancellations, or permission prompt rejections prior to file buffer loading.

Crash & Stability Health

Version	Active Users	Sessions	Engagement Rate	Crash-Free Rate	Release Status
1.3.07	136	152	86.8%	**100.0%**	**Production Baseline** (95.1% scale)
1.3.08	5	13	**53.8%**	**100.0%**	**Staged Rollout Alert** (3.5% scale)
1.3.06	2	4	100.0%	100.0%	Legacy Version

■ New Build / Staged Rollout Alert Section

CRITICAL REGRESSION WARNING: App Version 1.3.08

- **Production Baseline (`1.3.07`)**: Engagement Rate = **86.8%** across 136 users.
- **Staged Release (`1.3.08`)**: Engagement Rate dropped to **53.8%** across 5 users (13 sessions).
- **Delta**: **-33.0% engagement degradation**.
- **Architectural Assessment**: Although zero crashes have been reported on `1.3.08`, the steep decline in engagement suggests potential UI freeze, non-responsive player controls, or hidden background exceptions during audio rendering. **Halt staged rollout immediately pending build verification.**

Top In-App Events & Feature Telemetry

Event Name	Event Count	Total Users	Events / User	Feature Analysis & Telemetry Assessment
`player_pitch_changed`	5,706	10	**570.6**	■ **TELEMETRY BUG**: Un-debounced UI Slider (firing event per pixel moved).
`player_tempo_changed`	2,046	9	**227.3**	■ **TELEMETRY BUG**: Continuous slider firing without debounce threshold.
`audio_sessions`	1,265	75	17.3	Core audio engine playback sessions created.
`user_engagement`	1,020	143	7.1	GA4 auto-tracked active engagement event.
`separation_play_started`	571	47	12.4	Active audio stem processing & playback execution.
`player_stem_muted`	209	5	41.8	High feature adoption among core power users.
`player_stem_balance_changed`	132	1	**132.0**	Un-debounced pan balance slider bug.

Geographic Distribution

Country	Active Users	Sessions	% Total User Base	Strategic Geo Assessment
South Korea	110	126	**76.9%**	Primary market driving current viral acquisition growth.
Iran	7	9	4.9%	Secondary market.
Bangladesh	5	13	3.5%	Organic long-tail traffic.

India	4	2	2.8%	Low session frequency.
Japan	3	3	2.1%	High-value monetization target market.
United States	3	2	2.1%	Core tier-1 expansion opportunity.
Other (4 countries)	6	8	4.2%	Combined long-tail traffic.

Prioritized Strategic Action Matrix

Priority	Category	Action Item	Technical / Architectural Specification	Owner
P0	**Telemetry**	**Fix Slider Debouncing**	Implement 300ms–500ms debounce buffer on UI sliders (`player_pitch_changed`, `player_tempo_changed`, `player_stem_balance_changed`) to eliminate analytics event spam.	Mobile Eng
P0	**Stability**	**Halt v1.3.08 Rollout**	Pause Apple TestFlight/App Store staged rollout of `1.3.08`. Investigate engagement drop (53.8%) via local UI performance testing.	Product / QA
P1	**Product**	**Optimize Import Funnel**	Add explicit user error logging (`import_error_permission`, `import_error_format`) to resolve 37.5% drop-off between file selection and processing start.	iOS Team
P1	**Growth**	**Capitalize on SK Market**	Localization optimization for Korean App Store (KOR ASO, localized metadata) and introduce paid tier while viral momentum is high.	Growth / ASO
P2	**Analytics**	**Fix Cohort Attribution**	Resolve GA4 (`not set`) user cohort anomalies in session start payloads by updating GA4 SDK initialization logic.	Mobile Arch

2. Vocal Cleaner (`com.neuralSound.audiocleaner`)

App Header & Health Score

- **GA4 Property ID**: `533563138` | **Stream ID**: `15407667671`
- **Overall Health Score**: **4.2 / 10**
- **Score Rationale**: Severe stability vulnerabilities (**66.67% crash-free rate**), lack of organic new user acquisition (`0 New Users` yesterday), and extreme StoreKit purchase execution friction (5 failed purchase events out of 12 attempts). The app maintains value due to power-user engagement duration (~58.8 min average session length) and active subscription renewals.

■ Multi-Horizon Growth & Historical Scale Comparison

Horizon	Active Users (AU)	New Users (NU)	Total Sessions	Engagement Rate	User Base Growth Velocity Analysis
Yesterday (Latest 24h)	3	0	17	88.2%	**Flat Acquisition**: Driven exclusively by returning power users.
Past 7 Days	11	7	65	73.8%	**Low Scale**: Micro-scale traffic with episodic user acquisition.
Past 30 Days	27	27	94	69.1%	**Static Lifecycle**: Entire user base established within last 30 days.
Past Year (365 Days)	27	27	94	69.1%	**Static Lifetime Boundary**.
All-Time / Lifetime	27	27	94	69.1%	**Lifetime Scale Baseline**.

```
[Lifetime AU: 27] ➔ [30D AU: 27 (100%)] ➔ [7D AU: 11 (40.7%)] ➔ [24h AU: 3 (11.1%)]
```

Trajectory & Retention Analysis

- **Scale Stagnation**: The application lacks top-of-funnel reach (27 total lifetime users).
- **High Session Intensity**: Despite the tiny user count, yesterday's 3 users generated **17 sessions** (5.66 sessions/user/day) and accumulated **59,937 seconds (~16.6 hours)** of total engagement time.

Old vs. New User Cohort Deep-Dive

User Cohort Type	Active Users	Total Sessions	Avg Session Duration	Engagement Rate	Screen / Page Views	Key Architectural Observation

New Users	0	0	0.0 sec	0.0%	0	**Acquisition Deficit**: Zero new user acquisitions in last 24h.
Returning Users	3	17	**3,531.1 sec (~58.8 min)**	88.2%	101	**Heavy Power Users**: Deep, persistent background audio processing.

Cohort Insights

- **Extreme Usage Profile**: Returning users maintain average sessions of **nearly one hour (3,531.1s)**, viewing **5.94 screens per session**. This usage pattern points to utility as a specialized workstation application (e.g., long-form audio denoiser processing long recordings).

7-Day Velocity & Trajectory

Date	Active Users	New Users	Sessions	Engagement Rate	Contextual Trend Analysis
2026-08-15	3	0	11	63.6%	Steady returning user usage.
2026-08-16	2	0	9	66.7%	Flat traffic volume.
2026-08-17	1	0	3	100.0%	Single power user activity.
2026-08-18	2	1	7	85.7%	Isolated new user acquisition.
2026-08-19	8	6	17	58.8%	Brief traffic spike; low engagement on acquisition.
2026-08-20	1	0	3	66.7%	Regression to baseline.
2026-08-21	3	0	17	**88.2%**	High session frequency return (17 sessions across 3 users).

User Funnel Diagrams

Recording & Audio Denoise Processing Flow

```
[2 Users] vc_record_open (6 events) ■ ▼ (100.0% Permission Granted) [2 Users] vc_record_permission_result (6 events) ■ ▼ (100.0% Capture Start) [2 Users] vc_record_start (8 events) ■ ▼ [2 Users] vc_record_pause (9 events) ■ ▼ [2 Users] vc_record_done (4 events) ■ ■■■■ [2 Users] vc_record_denoise_tap (4 events) ■■■ Core Processing Applied ■ ■■■■ [1 User] vc_record_denoise_blocked (2 events) ■■■ ■ Feature Gated / Paywall Triggered
```

Paywall & StoreKit Purchase Funnel

```
[1 User] paywall_viewed (16 events) ■ ▼ [1 User] paywall_plan_selected (2 events) ■ ▼ [1 User] purchase_started (12 events) / subscribe_tapped (12 events) ■ ▼ (83.3% Approval) [1 User] purchase_approved (10 events) ■ ■■■■ [1 User] purchase_succeeded / purchase (6 events) ■■■ Success Conversion ■ ■■■■ [1 User] purchase_failed (5 events) / StoreKit Fail (2 events) ■■■ ■ Purchase Friction
```

- **Monetization Bottleneck**: A single user was forced to attempt purchasing **12 times** to achieve **6 successful transactions**, logging **5 failures** and **2 explicit StoreKit errors**. This friction indicates StoreKit 2 retry loop issues or Sandbox/Production receipt validation delays.

Crash & Stability Health

Version	Active Users	Sessions	Engagement Rate	Crash-Free User Rate	Technical Risk Assessment
1.0.01	3	17	88.2%	**66.67%**	■ **CRITICAL RISK**: 1 out of 3 daily users experienced an app crash (`app_exception`).

■ New Build / Staged Rollout Alert Section

CRITICAL STABILITY ALERT: App Version 1.0.01

- **Current Production Build**: `1.0.01`
- **Crash-Free User Rate**: **66.67%** (1 fatal app crash logged via `app_exception` across 3 active users).
- **Architectural Assessment**: Memory leaks likely occur during long-running background DSP denoise tasks (accounting for average session durations of ~58.8 minutes). **Immediate patch (v1.0.02) required.**

Top In-App Events & Feature Telemetry

Event Name	Event Count	Total Users	Events / User	Feature Analysis & Telemetry Assessment
`user_engagement`	167	3	55.7	High persistent engagement duration.
`screen_view`	101	2	50.5	Active UI navigation loops.
`app_store_subscription_renew`	35	2	17.5	Subscription renewals firing active signals.
`paywall_viewed`	16	1	16.0	Repeated paywall impressions shown to active user.
`purchase_started`	12	1	12.0	High buy intent coupled with execution retries.
`purchase_succeeded`	6	1	6.0	Monetized conversion events.

`purchase_failed`	5	1	5.0	StoreKit error state logged.
`app_exception`	1	1	1.0	Unhandled native runtime exception crash.

Geographic Distribution

Country	Active Users	Sessions	% Total User Base	Strategic Geo Assessment
Bangladesh	3	17	**100.0%**	**Geographic Isolation**: Entire active population resides in a single market.

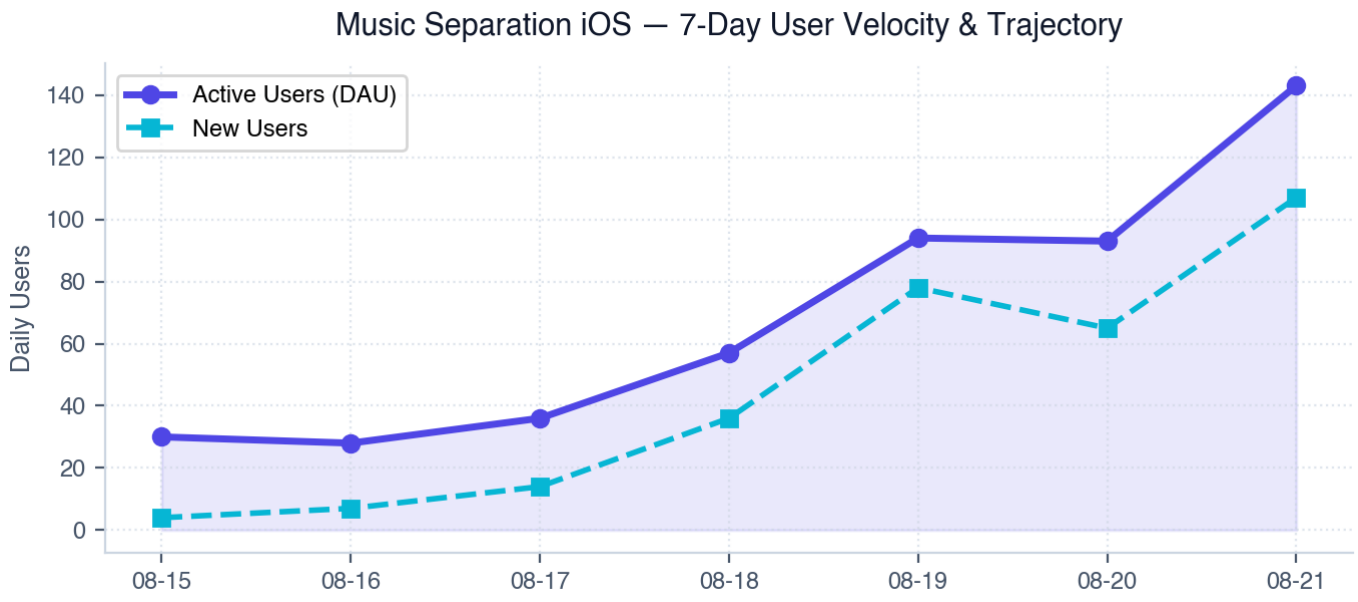
Prioritized Strategic Action Matrix

Priority	Category	Action Item	Technical / Architectural Specification	Owner
P0	**Engineering**	**Fix Crash Regression**	Debug background DSP audio denoise memory buffers (`app_exception`). Resolve memory leaks occurring during extended sessions.	iOS Core Eng
P0	**Monetization**	**Fix StoreKit 2 Retry Loop**	Overhaul StoreKit transaction listener to eliminate duplicate transaction prompts, transaction looping, and `purchase_failed` states.	Payment Eng
P1	**Growth**	**Kickstart Organic UA**	Launch ASO campaign (App Store metadata, targeted audio editing keywords) to resolve the acquisition shortfall (0 new users).	Growth Lead
P1	**Product**	**Paywall Gating Audit**	Audit `vc_record_denoise_blocked` events to ensure paywall triggers on denoise completion rather than blocking users during recording.	Product / UX

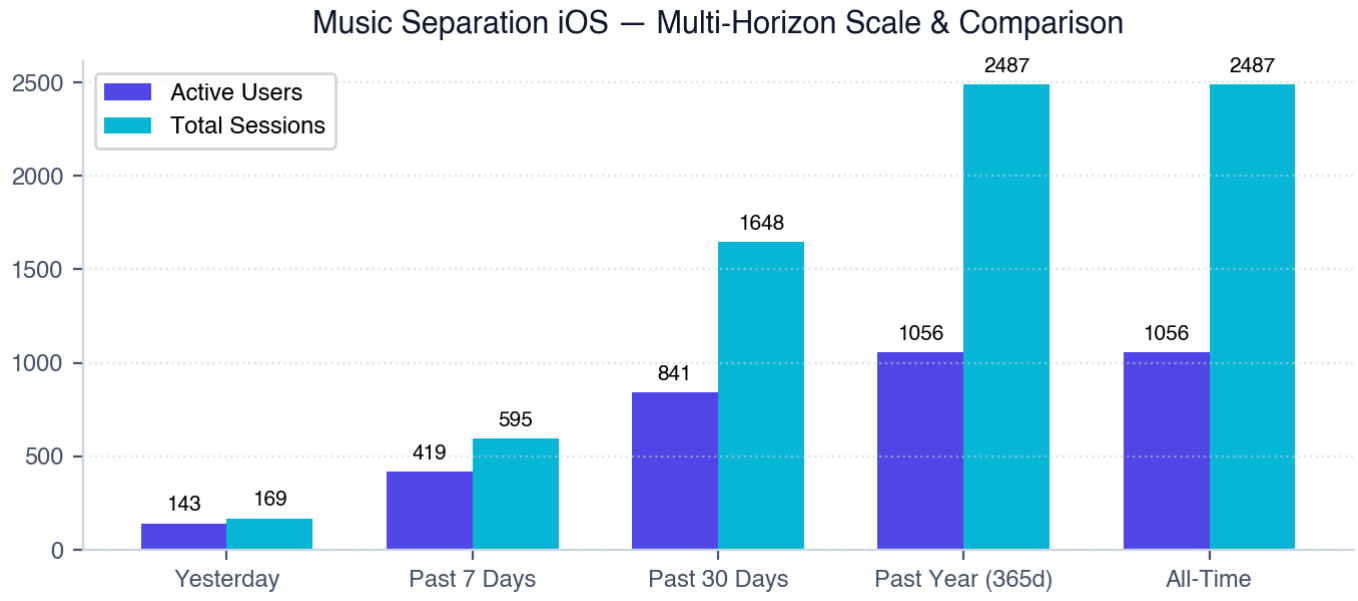
P2	**Analytics**	**Refine Purchase Logging**	Consolidate duplicate purchase telemetry (`purchase`, `purchase_succeeded`, `purchase_approved`) into a single GA4 `purchase` event.	Analytics

■ Visual Graphs & Performance Charts

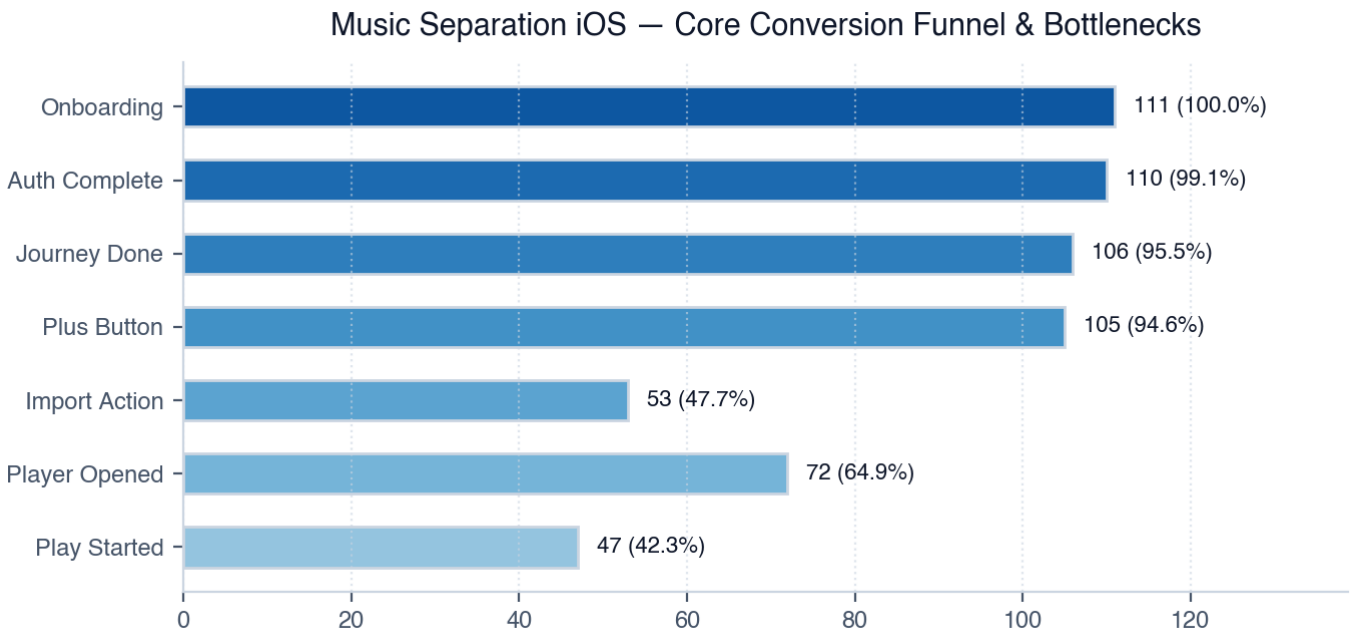
Music Separation Ios Trend



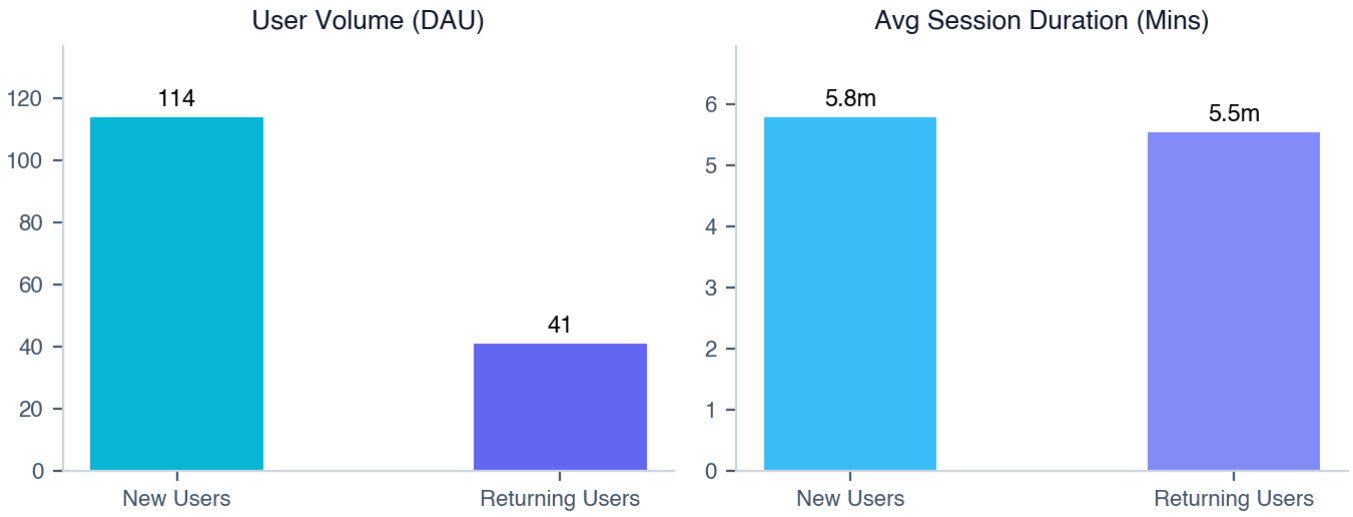
Music Separation Ios Horizons



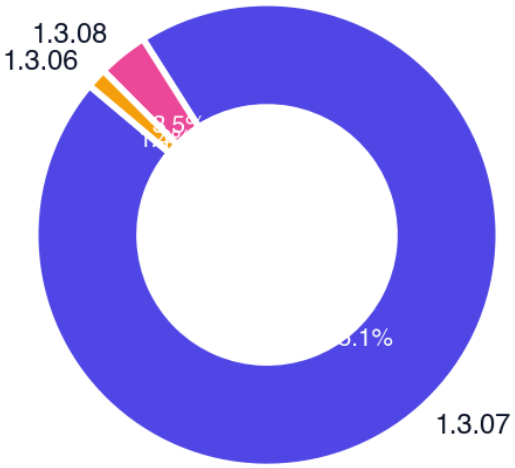
Music Separation Ios Funnel



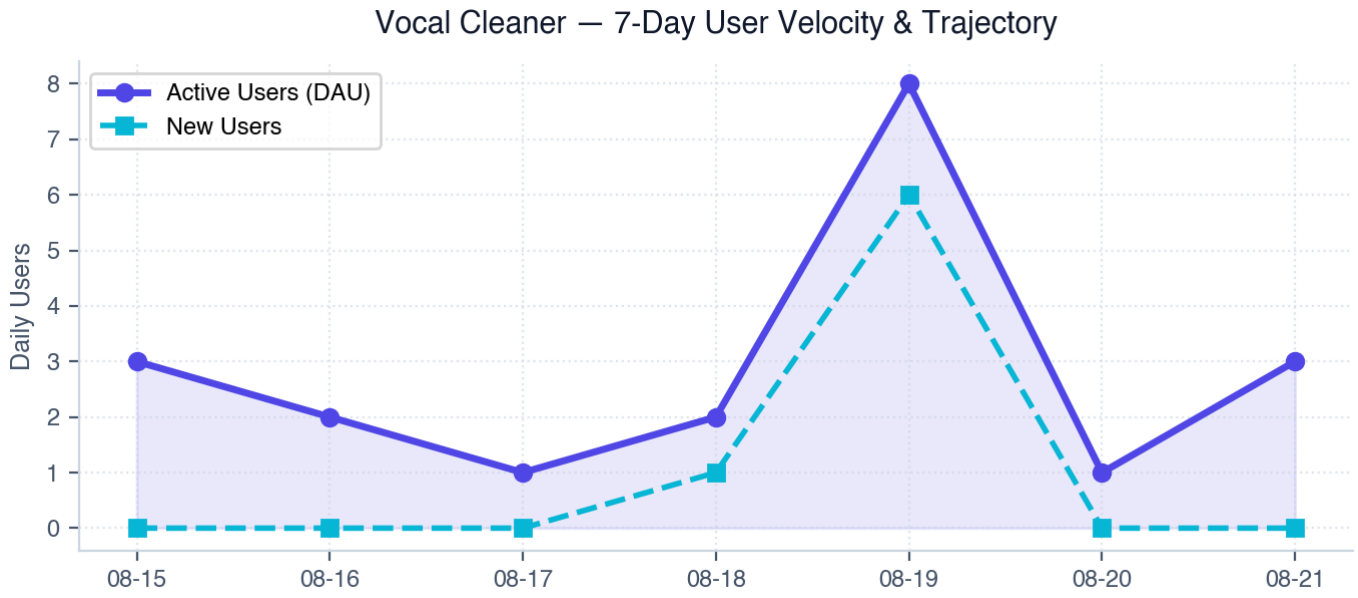
Music Separation Ios Cohort



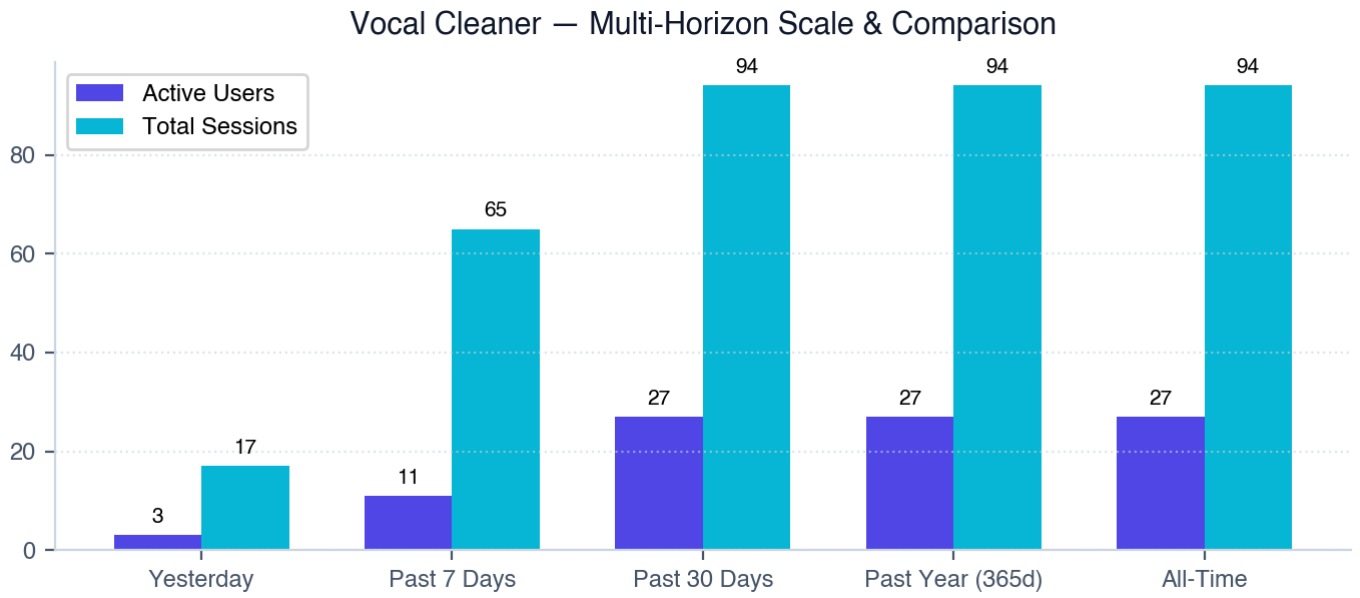
Music Separation Ios Version



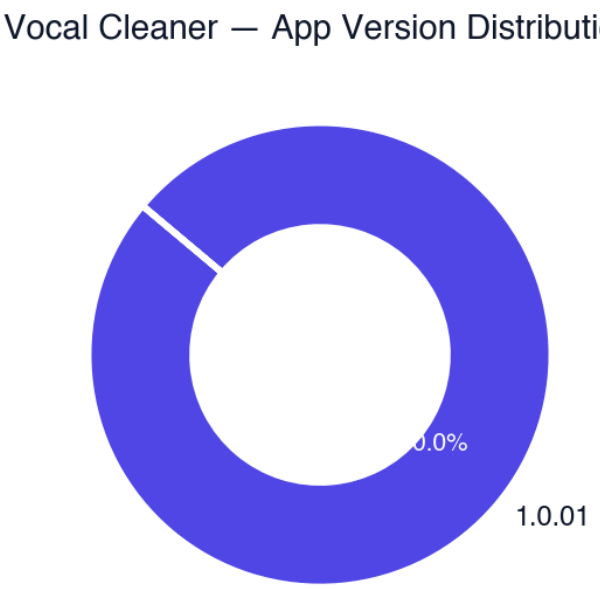
Vocal Cleaner Trend



Vocal Cleaner Horizons



Vocal Cleaner Version



Direct Console Links:

■ Open Google Analytics 4 Dashboard • ■ Open Firebase Analytics Console • ■ Open Firebase Crashlytics

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